Topic 1.13 - Function Model Selection and Assumption Articulation

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: ____________ Class: ____________ Date: ____________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use differences, ratios, rates, and graphical shape to compare candidate model families.
- Compute and interpret residuals and use residual patterns to evaluate fit.
- State assumptions, contextual domains, interpolation limits, and extrapolation risks in precise language.

Section A: Foundation and representation

Question 1

Pattern | No calculator

For equally spaced inputs, outputs are $11, 17, 23, 29, 35$. Select a model family and justify using differences.

Question 2

Pattern | No calculator

For equally spaced inputs, outputs are $2, 6, 18, 54, 162$. Select a model family and identify the growth factor.

Section A continued

Question 3

Construction | No calculator

Outputs are $4, 7, 12, 19, 28$ for inputs $0, 1, 2, 3, 4$. Select and construct a model.

Question 4

Residual | No calculator

An observed output is 83 and a model prediction is 79.5. Compute and interpret the residual.

Question 5

Residual analysis | No calculator

Model A residuals are $-1.2, -0.8, -0.1, 0.7, 1.4$; Model B residuals are $0.5, -0.4, 0.2, -0.3, 0.1$. Which is more defensible and why?

Section B: Application and reasoning

Question 6

Error analysis | No calculator

A model has $R^2 = 0.998$ but a U-shaped residual plot. Evaluate the model choice.

Question 7

Assumptions | No calculator

A school attendance model assumes enrollment and transport access remain stable. Explain why each assumption matters.

Question 8

Domain | No calculator

A data set records the number of defects in each manufactured item. Should the input domain be continuous or discrete? Explain.

Section C: Challenge and error analysis

Question 9

Model selection | No calculator

A population grows by roughly 8% per year for six years. Compare linear and exponential interpretations and select one if the percentage is stable.

Question 10

Challenge | No calculator

A chemical response rises rapidly and then approaches a stable ceiling. Explain why an unrestricted polynomial may be less plausible than a rational model with a horizontal asymptote.

Question 11

Investigation | No calculator

Two models fit sales data from months 1 through 6. At month 6, one has a large positive residual while earlier residuals are near zero. What should be investigated?

Question 12

Critique | No calculator

A student uses a model fitted to temperatures from 15°C to 30°C to predict at 100°C and reports six decimal places. Identify two weaknesses.

Section D: AP-style multiple choice

MCQ 1
No calculator

A residual of -6 means the model

- **A.** underpredicted by 6
- **B.** overpredicted by 6
- **C.** has slope -6
- **D.** is invalid

MCQ 2
No calculator

Which pattern most strongly supports an exponential model at equally spaced inputs?

- **A.** Constant first differences
- **B.** Constant second differences
- **C.** Constant ratios
- **D.** Constant residuals equal to 5

MCQ 3
No calculator

Which residual plot is most desirable?

- **A.** A clear upward line
- **B.** A U-shaped curve
- **C.** Small points randomly scattered around zero
- **D.** All residuals positive

MCQ 4
No calculator

Which statement is an assumption rather than a computed result?

- **A.** The residual is 2.3
- **B.** The market conditions remain similar next year
- **C.** The quadratic coefficient is -4
- **D.** The table contains eight points

AP-style free response 1

Calculator permitted - 6 points

A laboratory records the values below.

x	0	1	2	3	4	5
y	4.1	7.0	14.2	24.8	40.1	59.0

Quadratic predictions are $4, 7, 14, 25, 40, 59$ and exponential predictions are $4.8, 7.5, 11.7, 18.3, 28.6, 44.7$.

A. Compute all residuals for both models.

B. Select a model and justify using residual magnitude and pattern.

C. Use the selected model to predict at $x = 8$.

D. State one assumption and one limitation of that prediction.

AP-style free response 2

No calculator - 6 points

A quantity has values $80, 48, 28.8, 17.28, 10.368$ at years $0, 1, 2, 3, 4$.

A. Use numerical structure to select a model family.

B. Construct a model $Q(t)$.

C. Interpret the key parameter controlling change.

D. Evaluate a prediction at $t = 12$ and explain whether it should be trusted automatically.